

An Overview of the Private Data Collection Feature in Hyperledger Fabric

This article is for open source technology enthusiasts with prior experience in Hyperledger Fabric.



In cases where a group of organisations on a channel need to keep data private from other organisations on that channel, they have the option of creating a new channel comprising just the organisations that need access to the data. However, creating separate channels in each of these cases creates additional administrative overheads like maintaining chaincode versions, policies, MSPs, etc, and doesn't allow for use cases that require all channel participants to see a transaction while keeping a portion of the data private. To fulfil this last requirement, in v1.2, Hyperledger Fabric offers the ability to create private data collections, which allow a defined subset of organisations on a channel the ability to endorse, commit, or query private data without having to create a separate channel.

Private data collection

A collection is the combination of two elements:

- The actual private data, sent peer-to-peer using the gossip protocol to only the organisation(s) authorised to see it. This data is stored in a private database on the peer. The ordering service is not involved here and does not see the private data. Note that setting up gossip requires setting up anchor peers in order to bootstrap cross-organisation communication.
- A hash of that data, which is endorsed, ordered, and written to the ledgers of every peer on the channel. The hash serves as evidence of the transaction and is used for state validation, and can be used for audit purposes.

Private data collection definition

A collection definition contains one or more collections, each having a policy definition listing the organisations

in the collection, as well as properties used to control the dissemination of private data at endorsement time and optionally, whether the data will be purged.

Collection definitions are composed of six properties.

name: The name of the collection.

policy: The private data collection distribution policy defines which of the organisations' peers are allowed to persist the collection data expressed using the signature policy syntax, with each member being included in an OR signature policy list. To support read/write transactions, the private data distribution policy must define a broader set of organisations than the chaincode endorsement policy, as peers must have the private data in order to endorse proposed transactions.

requiredPeerCount: This is the minimum number of peers (across authorised organisations) that each endorsing peer must successfully disseminate private data to, before the peer signs the endorsement and returns the proposal response back to the client. Requiring dissemination as a condition of endorsement will ensure that private data is available in the network even if the endorsing peer(s) become unavailable. When the *requiredPeerCount* is 0, it means that no distribution is required, but there may be some distribution if *maxPeerCount* is greater than zero.

maxPeerCount: For data redundancy purposes, this count is the maximum number of other peers (across authorised organisations) that each endorsing peer will attempt to distribute the private data to. If an endorsing peer becomes unavailable between endorsement time and commit time, other peers that are collection members but who did not yet receive the private data at endorsement time, will be able to pull the